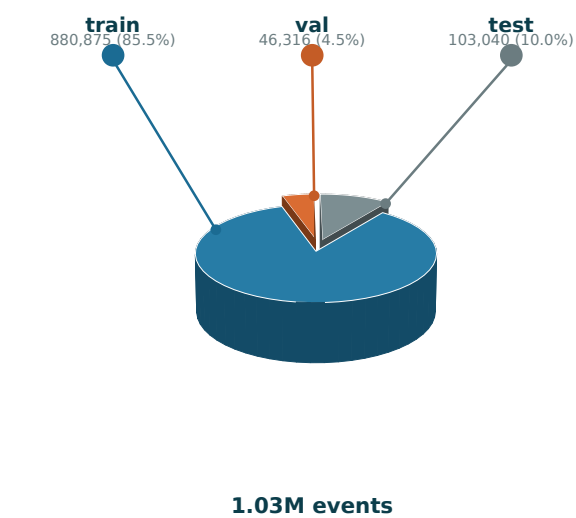
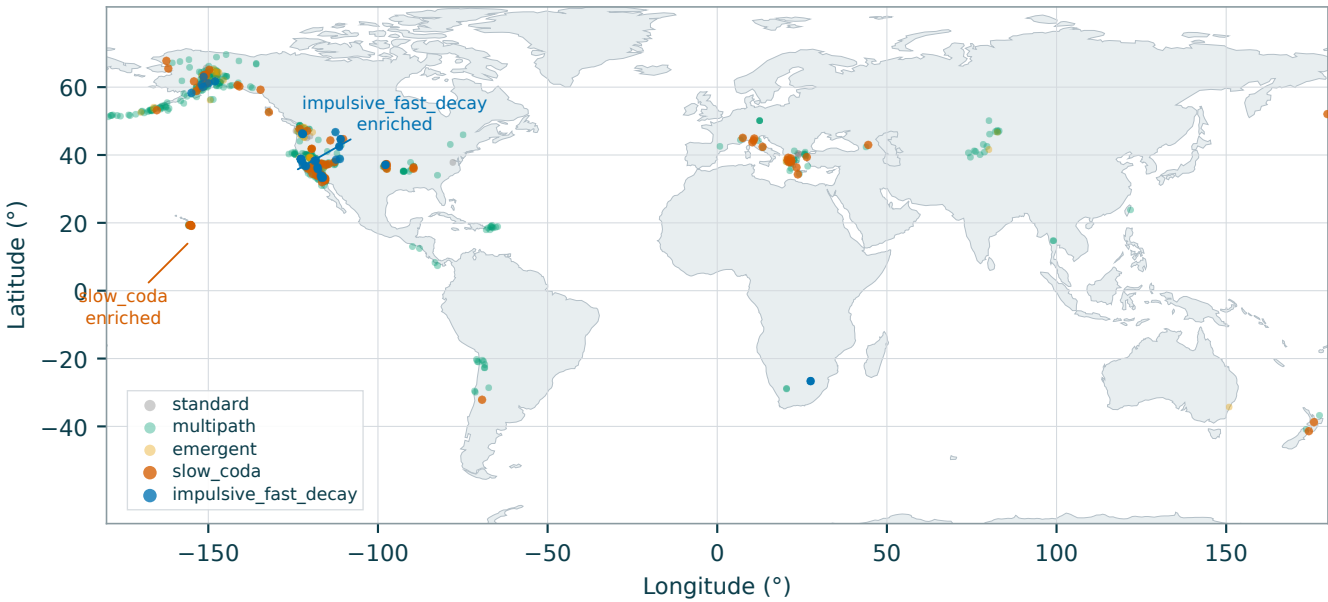


Data-driven physics discovery: waveform morphology clusters linked to attenuation structure

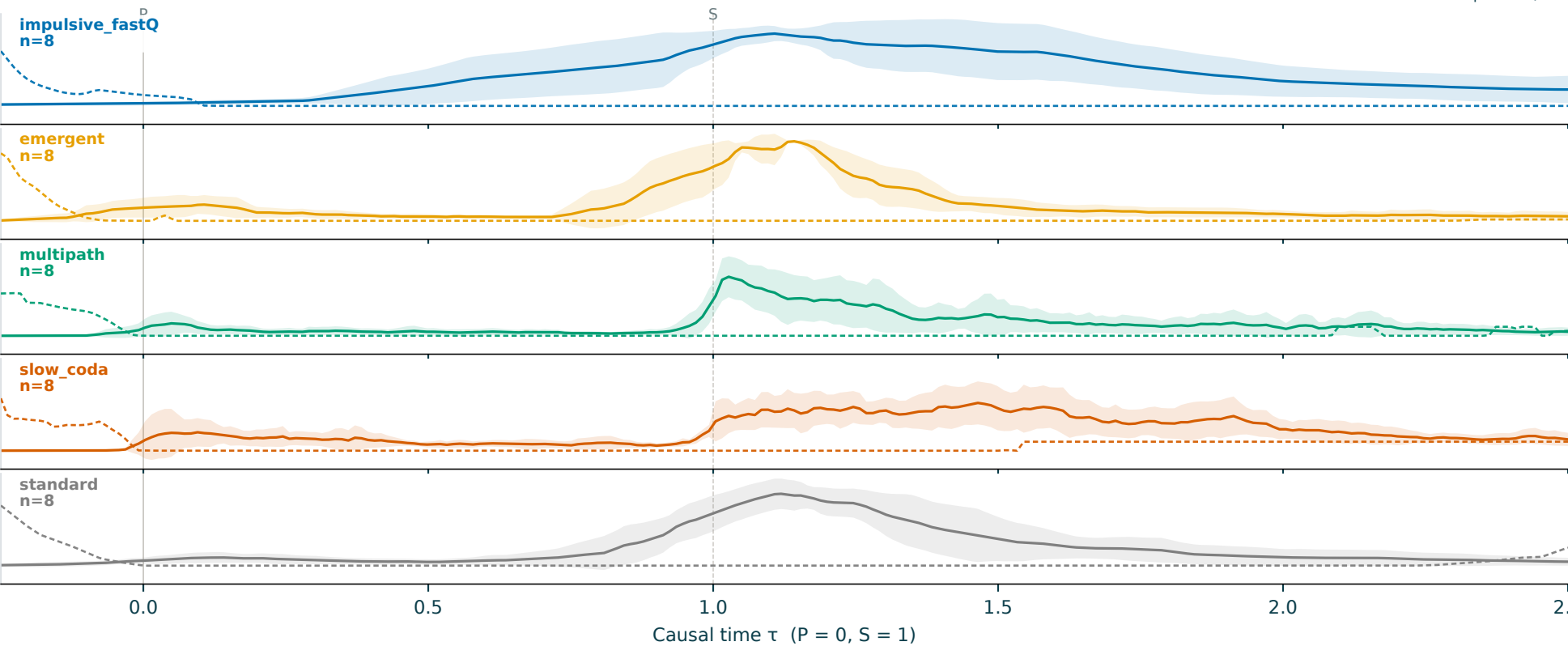
a STEAD dataset split



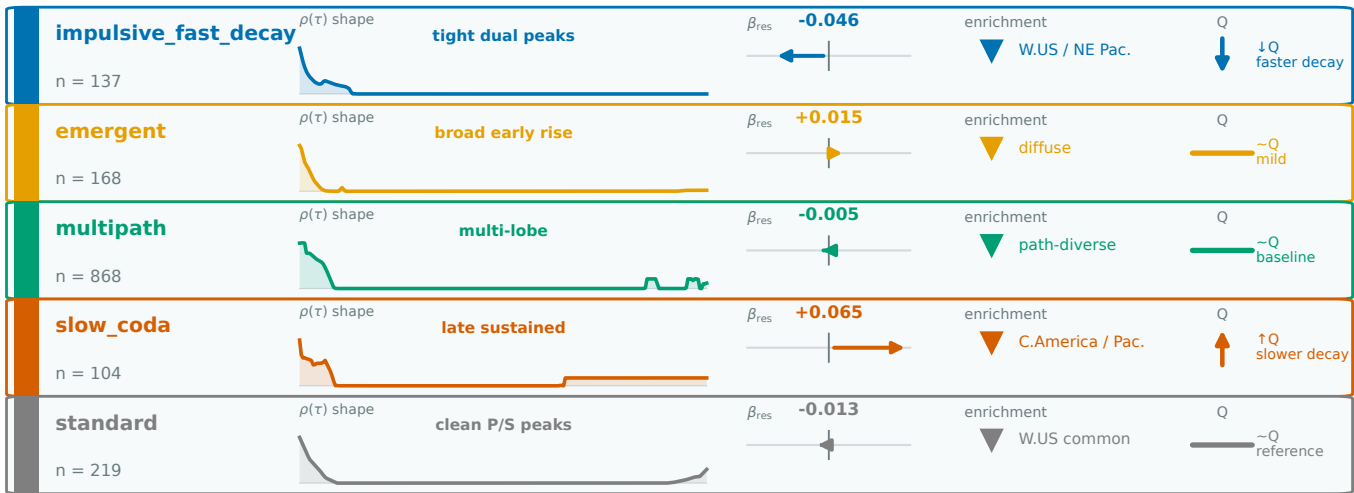
b Geographic distribution — regional enrichment of classes



c Normalised waveforms and $\rho(\tau)$ for five morphology classes



d Morphology-class characters



e Distance-detrended coda residual

